Book 1 — Lecture 2 — Exercise 8

Velocity, Speed, and Acceleration for Four Plane Motions

Prompt. Calculate the velocity, speed, and acceleration for each of the following position vectors. If you have graphing software, plot each position vector, each velocity vector, and each acceleration vector:

$$\begin{aligned}
&\text{(i)} && \mathbf{r}(t) = \big(\cos(\omega t),\, e^{\omega t}\big),\\
&\text{(ii)} && \mathbf{r}(t) = \big(\cos(\omega t - \phi),\, \sin(\omega t - \phi)\big),\\
&\text{(iii)} && \mathbf{r}(t) = \big(c\cos^3 t,\, c\sin^3 t\big),\\
&\text{(iv)} && \mathbf{r}(t) = \big(c(t - \sin t),\, c(1 - \cos t)\big).
\end{aligned}$$

[1]

Throughout, $\mathbf{v} = \dfrac{d\mathbf{r}}{dt}$ and $\mathbf{a} = \dfrac{d\mathbf{v}}{dt}$. Speed is $|\mathbf{v}|$.

(i) $\mathbf{r} = \big(\cos(\omega t),\, e^{\omega t}\big)$

Components $x = \cos(\omega t)$, $y = e^{\omega t}$. Then

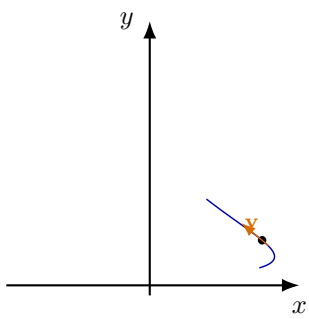
$$v_x = -\omega \sin(\omega t), \qquad v_y = \omega e^{\omega t}, \qquad |\mathbf{v}| = |\omega|\sqrt{\sin^2(\omega t) + e^{2\omega t}}.$$

Also

$$a_x = -\omega^2 \cos(\omega t), \qquad a_y = \omega^2 e^{\omega t}.$$

The path is *not* a circle; speed grows roughly like $|\omega| e^{\omega t}$ for $\omega t \gg 1$.

Figure (i). The sketch uses $\omega = 1$ and plots $x = \cos t$, $y = e^t$ for $t \in [-0.5, 1.1]$ (the vertical scale is compressed so the curve fits on the page).



[1] Here ω, ϕ, c are constants and t is time. The exponential is $e^{\omega t}$ (not $\exp(\omega)$ at a single point). The cycloid uses $\cos t$ (a common slip is "`cost`" for $\cos t$). Plotting is optional but helps intuition: $\mathbf{v}$ is tangent to the path and $|\mathbf{v}|$ is the scalar speed.

Tangent $\mathbf{v}$ at a sample time; $\mathbf{a}$ points up–right here and is not drawn to scale with $\mathbf{v}$.

(ii) $\mathbf{r} = \big(\cos(\omega t - \phi),\, \sin(\omega t - \phi)\big)$

Let $\theta = \omega t - \phi$. Then $x = \cos\theta$, $y = \sin\theta$, and $\dot{\theta} = \omega$. By the chain rule,

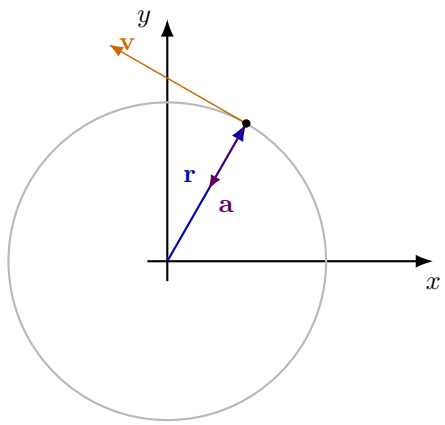
$$v_x = -\omega \sin\theta, \qquad v_y = \omega\cos\theta, \qquad |\mathbf{v}| = |\omega|\sqrt{\sin^2\theta + \cos^2\theta} = |\omega|.$$

Differentiate again:

$$a_x = -\omega^2\cos\theta, \qquad a_y = -\omega^2\sin\theta = -\omega^2\,\mathbf{r}.$$

Uniform circular motion: constant speed $|\omega|$ and centripetal acceleration of magnitude $\omega^2 R$ with $R = 1$ here.

Figure (ii). Unit circle with $\mathbf{r}$ outward, $\mathbf{v}$ tangent, and $\mathbf{a} = -\omega^2\mathbf{r}$ inward (illustration uses $\omega = 1$ for arrow lengths).



(iii) $\mathbf{r} = \big(c\cos^3 t,\, c\sin^3 t\big)$ (astroid / hypocycloid track)

Write $x = c\cos^3 t$, $y = c\sin^3 t$. Then

$$v_x = -3c\cos^2 t\sin t, \qquad v_y = 3c\sin^2 t\cos t.$$

Factor:

$$v_x^2 + v_y^2 = 9c^2\cos^2 t\sin^2 t(\cos^2 t + \sin^2 t) = 9c^2\cos^2 t\sin^2 t = \frac{9c^2}{4}\sin^2(2t),$$

so

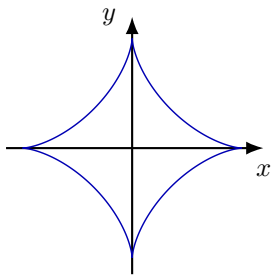
$$|\mathbf{v}| = \frac{3|c|}{2}\,|\sin(2t)|.$$

Differentiate v_x and v_y (product rule):

$$\begin{aligned} a_x &= -3c(-2\cos t\sin^2 t + \cos^3 t) = 3c\cos t(2\sin^2 t - \cos^2 t), \\ a_y &= 3c\sin t(2\cos^2 t - \sin^2 t). \end{aligned}$$

At multiples of $t = 0, \pi/2, \ldots$, the velocity vanishes (cusps of the projected curve in the plane).

Figure (iii). Trace with $c = 1$: $(\cos^3 t,\, \sin^3 t)$ for $0 \le t \le 2\pi$ (four symmetric cusps).



(iv) Cycloid $\mathbf{r} = \big(c(t - \sin t),\, c(1 - \cos t)\big)$

$$v_x = c(1 - \cos t), \qquad v_y = c \sin t.$$

Now

$$\begin{aligned} v_x^2 + v_y^2 &= c^2\big[(1 - \cos t)^2 + \sin^2 t\big] = c^2\big[1 - 2\cos t + \cos^2 t + \sin^2 t\big] \\ &= 2c^2(1 - \cos t) = 4c^2 \sin^2(t/2), \end{aligned}$$

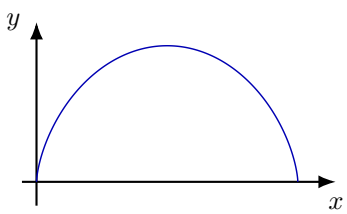
using $1 - \cos t = 2\sin^2(t/2)$. Hence

$$|\mathbf{v}| = 2|c|\,|\sin(t/2)|.$$

Finally,

$$a_x = c \sin t, \qquad a_y = c \cos t.$$

Figure (iv). One arch of the standard cycloid with $c = 1$: $(t - \sin t,\, 1 - \cos t)$ for $0 \le t \le 2\pi$.



Result. **(i)** $\mathbf{v} = (-\omega \sin \omega t,\, \omega e^{\omega t})$, $|\mathbf{v}| = |\omega|\sqrt{\sin^2 \omega t + e^{2\omega t}}$, $\mathbf{a} = (-\omega^2 \cos \omega t,\, \omega^2 e^{\omega t})$. **(ii)** $\mathbf{v} = (-\omega \sin(\omega t - \phi),\, \omega \cos(\omega t - \phi))$, $|\mathbf{v}| = |\omega|$, $\mathbf{a} = -\omega^2 \mathbf{r}$. **(iii)** $\mathbf{v} = (-3c\cos^2 t \sin t,\, 3c \sin^2 t \cos t)$, $|\mathbf{v}| = \dfrac{3|c|}{2}|\sin 2t|$, $\mathbf{a}$ as in the section above. **(iv)** $\mathbf{v} = c(1 - \cos t, \sin t)$, $|\mathbf{v}| = 2|c \sin(t/2)|$, $\mathbf{a} = c(\sin t, \cos t)$.

Example. Graphing: plot $\mathbf{r}(t)$ as a parametric curve; attach arrows for $\mathbf{v}(t)$ and $\mathbf{a}(t)$ at a few sample times (e.g. Desmos, GeoGebra, Matplotlib, or a CAS). For (ii) the hodograph of $\mathbf{v}$ is a circle; for (iv) the rolling-wheel interpretation matches the cycloid form.